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(54) **METHOD FOR MELTING DIRECT REDUCED IRON, SOLID IRON AND METHOD FOR PRODUCING THE SAME, MATERIAL FOR CIVIL ENGINEERING AND CONSTRUCTION AND METHOD FOR PRODUCING THE SAME, AND SYSTEM FOR MELTING DIRECT REDUCED IRON**

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(57) **ABSTRACT**

A technology for melting direct reduced iron while efficiently removing a gangue portion from the direct reduced iron, including obtaining a direct reduced iron by bringing iron ore or a mixture thereof and composition adjusting material together with a reducing material under heating; melting the direct reduced iron to obtain molten iron, and removing a slag outside the induction melting furnace; and optionally refining the molten iron. A charging temperature of melting the direct reduced iron ranges from after finishing direct reduction to atmospheric temperature; and melting includes blowing gas into the molten iron for a limited time of or throughout melting, and optionally includes one or more steps: 1) adding an adjuster for adjusting components of the slag, 2) supplying heat to the slag from a heat source disposed above the induction melting furnace, and 3) supplying one or more reducing solids and/or one or more reducing gases.

FIG. 1

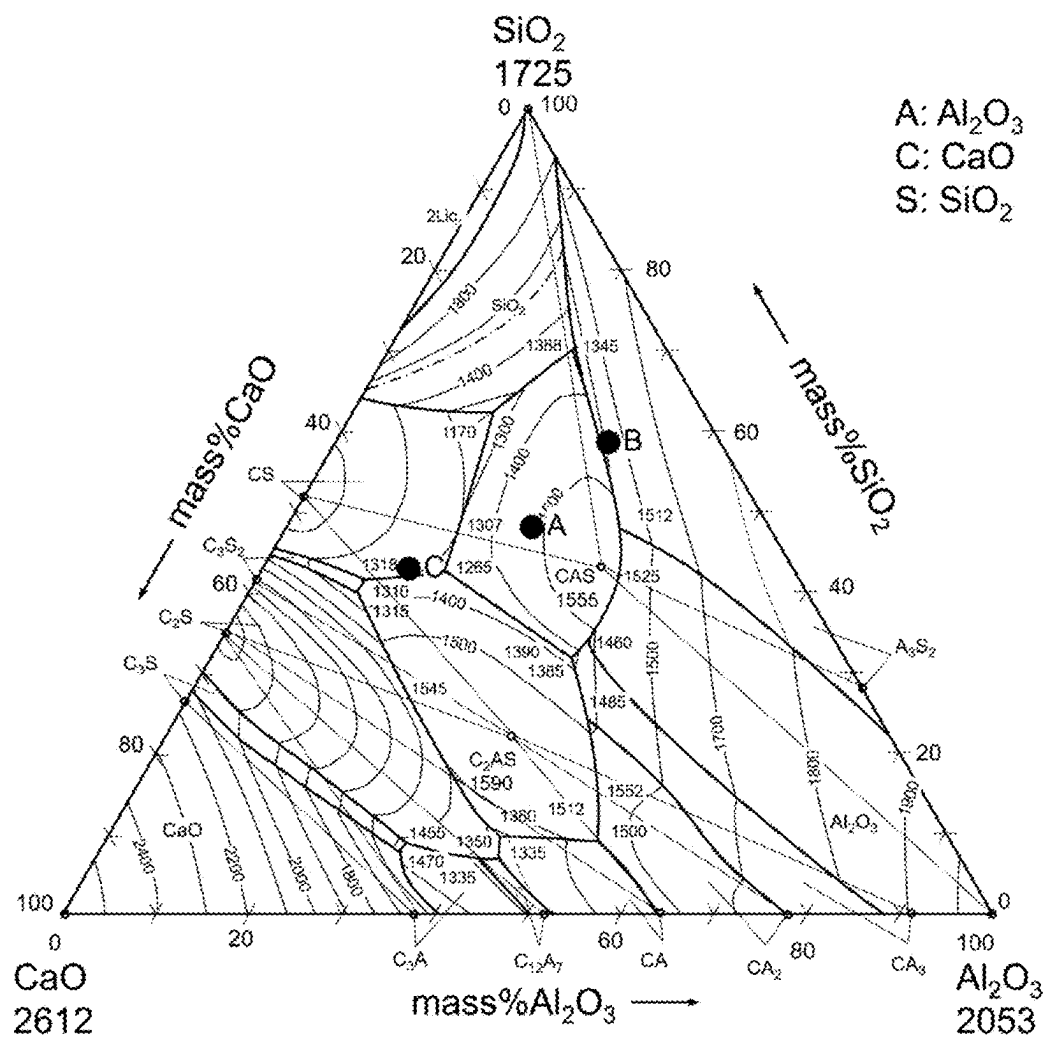
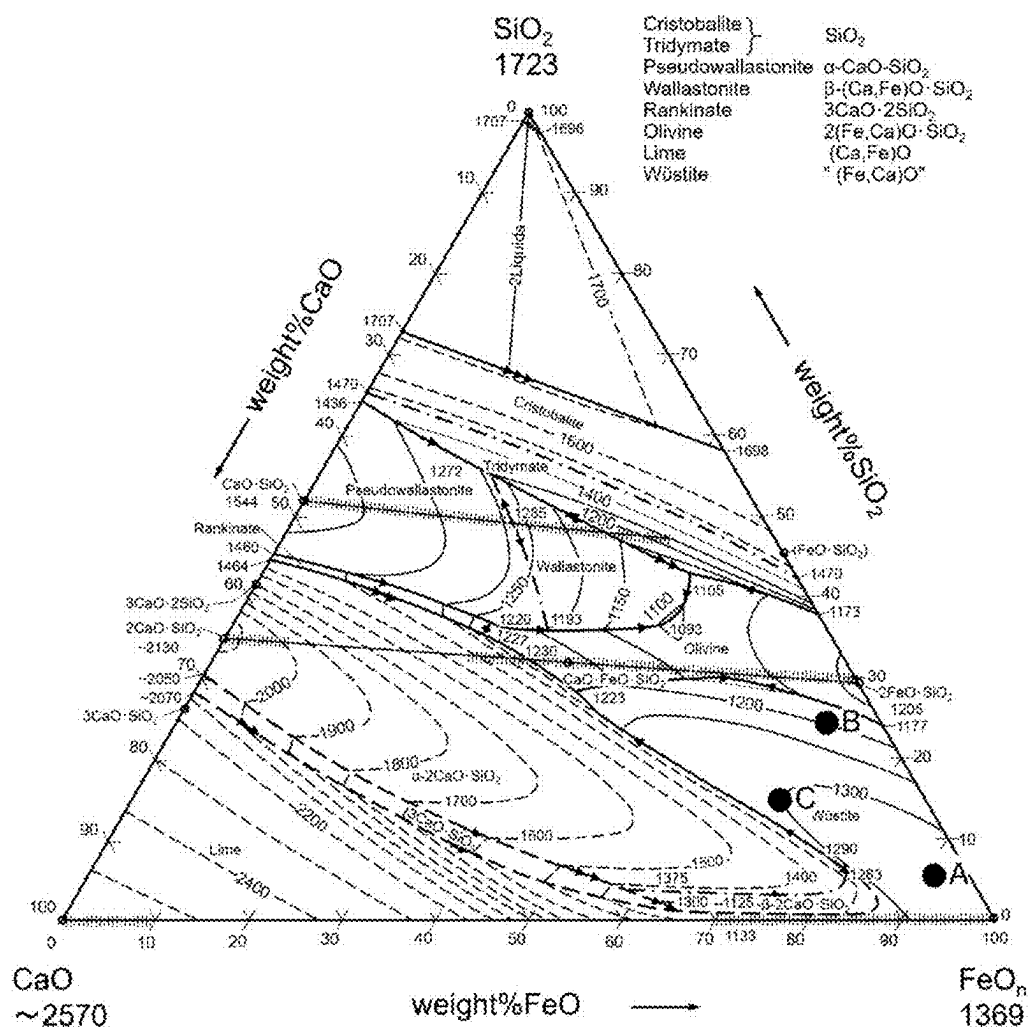


FIG. 2



**METHOD FOR MELTING DIRECT
REDUCED IRON, SOLID IRON AND
METHOD FOR PRODUCING THE SAME,
MATERIAL FOR CIVIL ENGINEERING AND
CONSTRUCTION AND METHOD FOR
PRODUCING THE SAME, AND SYSTEM
FOR MELTING DIRECT REDUCED IRON**

TECHNICAL FIELD

[0001] The present invention relates to a method for melting direct reduced iron including removing gangue contained in the direct reduced iron; solid iron and a method for producing the same each using such a melting method; a material for civil engineering and construction and a method for producing the same; and a system for melting direct reduced iron.

BACKGROUND ART

[0002] In recent years, there has been a growing demand for increased use of cold iron sources (scrap) in the steel industry. To build a recycling society, recycling iron sources is essential. Moreover, increasing the use of scrap is essential from the perspective of preventing global warming and responding to the demand for reducing CO₂ emissions. The amount of scrap used can reduce CO₂ emissions because the production process for the scrap does not require a reduction step unlike with iron ore which is iron oxide (Fe₂O₃). Thus, the amount of cold iron sources used is increasing more and more.

[0003] The blast furnace-converter method is a steelmaking process that includes supplying iron ore (Fe₂O₃) as a raw material into a blast furnace together with coke (a carbon source) as a reducing material, to obtain molten pig iron with a C concentration of approximately 4.5 to 5 mass %, and then supplying the obtained molten pig iron into a converter to remove C, Si, and P, which are impurity components, through oxidation. When molten pig iron is produced in a blast furnace, a reduction process for iron ore and the like require approximately 500 kg of carbon source to produce 1 ton of molten pig iron, generating approximately 2 tons of CO₂ gas. Meanwhile, when molten steel is produced using iron scrap as a raw material, no carbon source is required for the reduction process for iron ore. When the energy needed to melt iron scraps is considered, replacing 1 ton of molten pig iron with 1 ton of iron scrap leads to a reduction of CO₂ emissions by about 1.5 tons. This shows that the amount of scraps used should be increased to reduce greenhouse gas emissions while maintaining production activity.

[0004] However, the tight supply-demand balance for iron scrap, especially high-grade iron scrap, which is essential for producing high-grade steel, has increased the need for reduced iron to replace scrap. Reduced iron is produced by reducing iron ore. In the case of reduced iron, it is not necessary to set the C concentration in the generated iron to a high level as in the blast furnace-converter method. Thus, since an excessive amount of carbon source is not used, CO₂ emissions can be reduced by approximately 0.2 tons per 1 ton of iron. Further, when a hydrogen gas or a hydrocarbon-based gas, such as a natural gas, is used as a reducing material instead of a carbon source, CO₂ emissions can be further reduced.

[0005] Meanwhile, iron ore to be a raw material of reduced iron has different ingredient compositions depend-

ing on where it is mined. The composition of iron ore is mainly evaluated based on the Fe content and gangue content. Table 1 shows examples of ingredient compositions of iron ore.

TABLE 1

	Ingredient Composition (mass %)		
	T•Fe	SiO ₂	Al ₂ O ₃
Iron ore 1	55.0	5.5	1.9
Iron ore 2	59.0	3.6	2.3
Iron ore 3	62.4	2.6	2.2

[0006] A Fe content is represented by the total iron content T•Fe (mass %) in the iron ore. The higher the value, the higher the Fe content, which means that the value of the raw material is high. The gangue content is represented by the total content of oxides other than Fe in the iron ore, including SiO₂ and Al₂O₃ as a major part and about 0.1 mass % CaO, MgO, and so on. The gangue components are removed as impurities in the smelting process of iron from iron ore. Thus, the higher the gangue content, the lower the Fe content, which means that the transport cost and smelting cost per unit of Fe content will increase.

[0007] Reduced iron produced using iron ore as a raw material has different properties, such as a metallization rate and composition, depending on the brand of the iron ore used, the type and unit consumption of a raw material composition adjusting material to be mixed, the type and unit consumption of a reducing material, a reduction temperature, and a scheme adopted for an apparatus for producing the reduced iron. Table 2 shows examples of the ingredient compositions of reduced iron.

TABLE 2

	Ingredient Composition (mass %)				
	T•Fe	M•Fe	SiO ₂	Al ₂ O ₃	CaO
Reduced iron A	88.8	68.5	1.8	1.0	1.0
Reduced iron B	87.5	78.9	4.0	2.0	0.9
Reduced iron C	79.0	61.5	5.0	1.9	5.0

[0008] In the production of reduced iron, CaO is typically added as a raw material composition adjusting material to form slag together with gangue contained in the iron ore, thus securing the strength of the reduced iron. Reduced iron with the minimum amount of slag and higher metallization rate will have a lower load on the following transport, melting, and refining steps. Meanwhile, since high-grade iron ore is required to produce such reduced iron, the cost of the raw material would increase, and the cost required for the reduction treatment performed to increase the metallization rate would also increase, for example, which are problematic. Therefore, there is a need to develop a process for efficiently separating slag from reduced iron, which was produced using low-cost, low-grade iron ore as a raw material and has a high slag content, to increase the content of metallic iron.

[0009] A method for producing reduced metal by removing gangue contained in such ore as slag has been proposed. For example, Patent Literatures 1 and 2 each propose a method for producing reduced metal by feeding a raw material containing metal-containing substance onto a solid

reducing material layer deposited on a hearth of a moving hearth furnace and then heating and reducing the material, and also bringing the material into a molten state at least once to separate slag from metal.

CITATION LIST

Patent Literature

- [0010] Patent Literature 1: JP2000-292069A
 [0011] Patent Literature 2: JP2004-204293A

Non Patent Literature

- [0012] Non Patent Literature 1: Slag Atlas, 2nd ed., Verlag Stahleisen GmbH, Duesseldorf, (1995), 105, 126.

SUMMARY OF INVENTION

Technical Problem

[0013] However, the conventional technologies have the following problems. The technology disclosed in Patent Literatures 1 and 2 is based on the premise of using a carbon-based solid raw material as a reducing material to separate slag from metal. Therefore, when using a reduction process performed with a hydrogen-based reducing material, which is expected to become the mainstream in the future from the perspective of reducing CO₂ emissions, the efficiency of separating slag from metal is considered to decrease. Specifically, the temperature of an atmosphere in the moving hearth furnace, such as a rotary hearth furnace, is typically about 1300° C. The reason that metal reduced at such a temperature melts is considered that the melting temperature of the metal decreases as the metal is carburized with the carbon-based solid raw material. Thus, when a reduction process is performed with a hydrogen-based reducing material, it is predicted that the melting temperature of metal will not decrease due to carburization, and thus, the liquid phase rate of the fed material will not increase, which means that separation between the metal and slag is less likely to proceed.

[0014] That is, gangue may be difficult to remove in a reduction step in some cases, making it necessary to separate the gangue from the produced reduced iron. Specifically, when reduced iron is melted in an induction melting furnace, it is necessary to prevent the solidification of slag produced from gangue.

[0015] The present invention has been made in view of the above circumstances, and it is an object of the present invention to propose a method for melting direct reduced iron including efficiently removing a gangue component from the direct reduced iron; and a system for melting direct reduced iron. Furthermore, it is another object of the present invention to provide high-purity solid iron and a method for producing the same, each using such a melting method; and a material for civil engineering and construction for which a by-product is utilized, and a method for producing such a material for civil engineering and construction.

Solution to Problem

[0016] The inventors have found that when melting direct reduced iron in an induction melting furnace, it is possible to suppress the solidification of slag and thus allow the slag to be easily removed to the outside of the furnace by promoting the heat transfer between the molten iron and slag

or by controlling the composition of the slag, and thus have achieved the present invention.

[0017] A method for melting direct reduced iron according to the present invention which advantageously solves the above problems includes a direct reduction step of obtaining a direct reduced iron by bringing iron ore or a mixture of iron ore and a composition adjusting material into contact with a reducing material under heating, a melting step of melting the direct reduced iron in an induction melting furnace to produce molten iron, a slag removal step of removing slag produced in the melting step to an outside of the induction melting furnace, and, optionally, a refining step of refining the molten iron obtained in the melting step, characterized in that a charging temperature of the direct reduced iron into the induction melting furnace in the melting step of melting the direct reduced iron is in a range from a temperature after finishing the direct reduction step to atmospheric temperature, and the melting step includes a first step of blowing gas into the molten iron for a limited time of or throughout the melting step, and, optionally, one or more steps selected from 1) a second step of adding a slag composition adjusting material, 2) a third step of supplying heat to the slag from a heat source disposed above the induction melting furnace, and 3) a fourth step of supplying one or more reducing solids and/or one or more reducing gases.

[0018] Note that the method for melting direct reduced iron according to the present invention is considered to have more preferable solution means as follows, for example.

[0019] (a) The first step includes, provided that a height $H(m)$ from a position of a gas supply nozzle for blowing the gas into the molten iron to a bath surface of the molten iron is represented by following Formula (1), blowing the gas into the molten iron so as to satisfy following Formula (2), where p_g represents a density (kg/m³) of the supplied gas, ρ_i represents a density (kg/m³) of the molten iron, Q represents a supply rate (Nm³/minute) of the gas, N represents the number (—) of gas supply nozzles, d represents a diameter (m) of each gas supply nozzle, D represents an inner diameter (m) of the induction furnace, W_{DRI} represents a weight (kg) of reduced iron supplied into the induction furnace, $(\%T.Fe)_{DRI}$ represents a total iron concentration (mass %) in the reduced iron, and h represents a height (m) from a bottom of the induction furnace to the position of the gas supply nozzle.

$$H = 1.27 \times W_{DRI} / (p_1 D^2) \times (\%T.Fe)_{DRI} / 100 - h \quad (1)$$

$$H > 0.18 \times (\rho_g Q^2 / \rho_i N^2 d^2)^{1/3} \quad (2)$$

[0020] (b) The second step includes adjusting a type and amount of the slag composition adjusting material added so that the composition of the slag produced in the melting step can achieve a basicity, which is a ratio of a CaO concentration (% CaO) to a SiO₂ concentration (% SiO₂) on a mass basis, in a range of 0.5 to 2.0 and an Al₂O₃ concentration (% Al₂O₃) in a range of 10 to 25 mass %.

[0021] (c) The fourth step includes adjusting a type and amount of the reducing solid(s) and/or reducing gas(es) to be supplied so that the composition of the slag produced in the melting step can achieve a total iron concentration (% T.Fe) of 20 mass % or less.

[0022] A method for producing solid iron according to the present invention which advantageously solves the above problems includes solidifying the molten iron obtained by any one of the above methods for melting direct reduced iron to thus obtain solid iron. In addition, solid iron according to the present invention is solid iron produced by such a production method, characterized in that a total iron concentration T.Fe is 93 mass % or more, and a total content of oxide components other than Fe is 3 mass % or less.

[0023] A method for producing a material for civil engineering and construction according to the present invention which advantageously solves the above problems includes a melting step of melting direct reduced iron in an induction melting furnace to obtain molten iron, a slag removal step of removing slag produced in the melting step to an outside of the melting furnace, and a cooling solidification step of cooling the slag removed in the slag removal step to solidify the slag into a raw material for civil engineering and construction, characterized in that a charging temperature of the direct reduced iron into the induction melting furnace in the melting step of melting the direct reduced iron is in a range from a temperature after finishing the direct reduction step to atmospheric temperature, and the melting step includes a first step of blowing gas into the molten iron for a limited time of or throughout the melting step, and, optionally, one or more steps selected from 1) a second step of adding a slag composition adjusting material and 2) a third step of supplying heat to the slag from a heat source disposed above the induction melting furnace. Furthermore, a material for civil engineering and construction according to the present invention is produced by such a production method, characterized in that a basicity, which is a ratio of a CaO concentration (% CaO) to a SiO₂ concentration (% SiO₂) on a mass basis, is in a range of 0.5 to 2.0, and an Al₂O₃ concentration (% Al₂O₃) is in a range of 10 to 25 mass %.

[0024] The system for melting direct reduced iron according to the present invention that advantageously solves the above problems includes a direct reduction furnace that is configured to obtain a direct reduced iron by bringing iron ore or a mixture of iron ore and a composition adjusting material into contact with a reducing material under heating, an induction melting furnace that is configured to obtain a molten iron by melting the direct reduced iron, and a slag removal mechanism that is configured to discharge a slag generated in the induction melting furnace to an outside of the melting furnace, and further optionally includes a refining facility that is configured to refine the molten iron melted in the induction melting furnace, characterized in that the direct reduced iron that is to be melted in the induction melting furnace is charged into the induction melting furnace at a temperature of a range from a temperature after finishing the direct reduction treatment to atmospheric temperature, the induction melting furnace includes a function of blowing a gas into the molten iron for a limited time of or throughout the melting of the direct reduced iron, and further optionally includes one or more functions selected from: 1) a function of adding a slag composition adjusting material, 2) a function of supplying heat to the slag from a heat source disposed above the induction melting furnace, and 3) a function of supplying one or more reducing solids and/or one or more reducing gases. In the system for melting direct reduced iron according to the present invention, it is considered that more preferable solutions can be brought

when the system includes two or more said induction melting furnaces with respect to one said direct reduction furnace.

Advantageous Effects of Invention

[0025] The method for melting direct reduced iron and the system for melting direct reduced iron according to the present invention includes, when melting direct reduced iron in an induction melting furnace, blowing gas into the molten iron and, optionally, performing at least one of supplying gas at an appropriate flow rate into the molten iron, controlling the composition of the produced slag, and supplying heat to the slag from a heat source disposed above the furnace. This ensures that the slag remains in a fluid state, allowing the slag to be separated while metallic iron contained in the direct reduced iron is melted. Particularly, since the direct reduced iron produced is able to be charged into the induction melting furnace while still maintaining a high temperature, the amount of time and energy required for melting can be reduced.

[0026] The method for producing solid iron and solid iron according to the present invention can produce high-purity solid iron by solidifying molten iron while separating slag from the molten iron, which is preferable. Further, the method for producing a material for civil engineering and construction and a material for civil engineering and construction according to the present invention can effectively utilize a by-product by recovering it, particularly by recovering it while adjusting the composition of its components.

BRIEF DESCRIPTION OF DRAWINGS

[0027] FIG. 1 is a graph obtained by plotting the slag composition obtained from the reduced iron composition described in Table 2 on an Al₂O₃—CaO—SiO₂ ternary phase diagram.

[0028] FIG. 2 is a graph obtained by plotting the slag composition obtained from the reduced iron composition described in Table 2 on a CaO—SiO₂—FeO ternary phase diagram.

DESCRIPTION OF EMBODIMENTS

[0029] Embodiments of the present invention will be specifically described. Note that the following embodiments only describe examples of a system and a method for embodying the technical idea of the present invention and do not specify the configuration of the present invention. That is, the technical idea of the present invention may be modified in various ways within the technical scope recited in the claims.

[0030] The inventors have conducted studies on the removal of gangue from reduced iron, on the premise of melting the reduced iron once in an induction melting furnace. When heating and melting reduced iron in an induction melting furnace, a metallic iron portion in the reduced iron can be directly and efficiently heated by an induced current. Meanwhile, since a slag component is not directly heated, the slag floating on the molten iron due to the difference in specific gravity solidifies, causing a problem that it is difficult to charge additional reduced iron.

First Embodiment

[0031] Through further studies on the condition suitable for melting reduced iron in an induction melting furnace and

separating a gangue component as slag, the inventors have discovered that it is possible to suppress the solidification of slag by blowing gas into a bath when melting reduced iron in an induction melting furnace, whereby the reduced iron can be efficiently melted and the slag can be separated therefrom. Further, it has been also found that performing at least one of supplying gas at an appropriate flow rate into the molten iron, controlling the composition of the produced slag, and supplying heat to the slag from a heat source disposed above the furnace allows the slag to be maintained in a fluid state, and thus the slag can be more preferably separated while metallic iron contained in the reduced iron is melted. Here, "slag in a fluid state" refers to a state in which the entire slag is red hot and high-temperature slag is constantly circulating.

[0032] Maintaining slag in a fluid state can prevent the slag, which would otherwise have solidified, from hindering an additional charge of reduced iron, effectively utilizing the volume of the induction melting furnace. Further, since the slag is flowing, the slag can be easily separated from the molten iron by allowing the slag to overflow from the furnace top or by removing the slag with a slag dragger, for example. To separate the slag, it is preferable to tilt the furnace body to limit the place to which the slag is to be removed from the viewpoint of handling the slag at a high temperature and repairing wear damage portions, for example.

[0033] The reduced iron can, for example, be obtained by bringing iron ore as the raw material or a mixture of iron ore and a composition adjusting material into contact with a reducing material while performing heating with a rotary hearth furnace or the like as a direct reduction furnace. As the composition adjusting material, there may be used, for example, quicklime containing CaO; as the reducing material, there may be used, for example, a carbon material powder as a solid carbon source, and a reducing gas such as H₂, CO, and CH₄. The temperature of the reduced iron that is to be charged into the induction melting furnace is in a range from a temperature after finishing the reduction treatment with the direct reduction furnace to atmospheric temperature. This is preferable because the amount of time and energy required for melting can be reduced due to the fact that a reduced iron having a high temperature is now charged into the induction melting furnace. Thus, it is preferred that the temperature of the reduced iron that is to be charged into the induction melting furnace be higher than atmospheric temperature, more preferably 300° C. or higher, even more preferably a temperature merely resulting from a minimum necessary temperature drop that is incurred by transporting the reduced iron from the direct reduction furnace to the induction melting furnace.

[0034] The first embodiment of the present invention has been obtained from the above studies and includes a direct reduction step of obtaining a direct reduced iron by bringing iron ore or a mixture of iron ore and a composition adjusting material into contact with a reducing material under heating, a melting step of melting the direct reduced iron in an induction melting furnace to obtain molten iron, and a slag removal step of removing a slag produced in the melting step to the outside of the induction melting furnace. The charging temperature of the direct reduced iron into the induction melting furnace in the melting step of melting the direct reduced iron is in a range from a temperature after finishing the direct reduction step to atmospheric temperature. The melting step includes a first step of blowing gas into the molten iron for a limited time of or throughout the melting step, and optionally one or more steps selected from 1) a

second step of adding a slag composition adjusting material, and 2) a third step of supplying heat to the slag from a heat source disposed above the induction melting furnace.

Second Embodiment

[0035] Next, the optimization of the gas blowing condition has been studied. When gas is supplied into the molten iron, the molten iron is agitated along with the rise of the gas. Thus, heat is transferred from the molten iron to the slag that was produced and risen. This increases the temperature of the slag to thus improve the fluidity of the slag. The gas supplied herein may be any type of gas that does not liquefy when supplied through a pipe. For example, since an oxidizing gas such as oxygen or carbon dioxide oxidizes the molten iron and thus decreases iron yield, it is more preferable to use an inert gas such as Ar or N₂.

[0036] It should be noted that if the amount of the supplied gas is too large, so-called blow-by will occur, i.e. the gas will pass up to the bath surface of the molten iron while remaining in a continuous phase. Such blow-by will cause a significant increase in the spitting of the molten iron and a reduction in the agitation of the molten iron by the supplied gas as well as the reaction efficiency, thus reducing the effect of heat transfer to the slag. The inventors have conducted various studies under different conditions and consequently found that provided that the height H(m) from the position of a gas supply nozzle for blowing a gas into the molten iron to the bath surface of the molten iron is represented by following Formula (1), it is possible to avoid blow-by by blowing the gas into the molten iron so as to satisfy following Formula (2).

$$H = 1.27 \times W_{DRI} / (\rho_1 D^2) \times (\% T \cdot Fe)_{DRI} / 100 - h \quad (1)$$

$$H > 0.18 \times (\rho_g Q^2 / \rho_1 N^2 d^2)^{1/3}, \quad (2)$$

[0037] where ρ_g represents the density (kg/m³) of the supplied gas,

[0038] ρ_l represents the density (kg/m³) of the molten iron,

[0039] Q represents the gas supply rate (Nm³/minute),

[0040] N represents the number (—) of gas supply nozzles,

[0041] d represents the diameter (m) of each gas supply nozzle,

[0042] D represents the inside diameter (m) of the induction furnace,

[0043] W_{DRI} represents the weight (kg) of reduced iron supplied into the induction furnace,

[0044] $(\% T \cdot Fe)_{DRI}$ represents the total iron concentration (mass %) in the reduced iron, and

[0045] h represents the height (m) from the bottom of the induction furnace to the position of the gas supply nozzle.

[0046] Meanwhile, if the amount of the gas blown in is too small, the agitation effect of the molten iron with the rise of the gas will be small and thus the amount of heat transferred from the molten iron to the slag will also be reduced. Therefore, it is preferable to supply gas of at least 0.01 Nm³/minute per 1 ton of molten iron.

[0047] The second embodiment of the present invention was obtained from the above studies and includes, in addition to the first embodiment, the first step of, provided that the height $H(m)$ from the position of the gas supply nozzle for blowing a gas into the molten iron to the bath surface of the molten iron is represented by Formula (1) above, blowing the gas into the molten iron so as to satisfy Formula (2) above.

Third to Fifth Embodiments

[0048] Next, the optimization of the composition of the slag has been studied. The slag fluidity largely depends on the slag composition. FIGS. 1 and 2 are obtained by converting the slag compositions contained in examples of the reduced iron composition shown in Table 2 such that the total contents of the three components Al_2O_3 — CaO — SiO_2 or CaO — SiO_2 — FeO are 100% and then plotting the results on their respective ternary phase diagrams. Note that each ternary phase diagram are referenced from Non Patent Literature 1. The FeO concentration in the reduced iron was calculated by multiplying the difference between the total iron concentration $T.Fe$ and the metallic iron concentration $M.Fe$ by 71.85 (the molecular weight of FeO)/55.85 (the atomic weight of Fe).

[0049] As can be seen from FIG. 1, there is a composition of slag with a melting temperature lower than that of the slag components contained in the reduced iron. However, although the slag having a composition with a high SiO_2 concentration has a low melting temperature, it has a high viscosity and low fluidity. Therefore, it is preferable to set the basicity (hereinafter referred to as slag basicity), which is the ratio of the CaO concentration (% CaO) to the SiO_2 concentration (% SiO_2) on a mass basis, to 0.5 or more. Further, from the viewpoint of lowering the melting temperature of the slag, the slag basicity should be 2.0 or less and the Al_2O_3 concentration in the slag should be in the range of 10 to 25 mass %. To adjust the slag composition, it is preferable to add a material containing one or more of CaO , SiO_2 and Al_2O_3 as a raw material composition adjusting material in the production of reduced iron or as a slag composition adjusting material in the melting of the reduced iron. The material containing CaO may be any one of limestone, slaked lime, quicklime, steelmaking slag, and so on. Note that limestone and slaked lime will undergo an endothermic reaction as they decompose, causing a temperature drop, while the steelmaking slag has a CaO concentration of about 40 to 50 mass % and may be added in a larger amount, increasing the amount of the slag produced when the reduced iron is melted, which is problematic. Therefore, it is preferable to use quicklime. The material containing SiO_2 can be any of silica stone, coal ash, and steelmaking slag. In addition, it is also possible to add metallic Si or silicon sludge to utilize SiO_2 produced through a reaction with an iron oxide component remaining in the reduced iron. As the material containing Al_2O_3 , it is possible to add natural stone, such as corundum or bauxite; metallic Al ; or aluminum dross to utilize Al_2O_3 produced by a reaction thereof with an iron oxide component remaining in the reduced iron.

[0050] The third embodiment of the present invention has been obtained from the above studies and includes, in addition to the first or second embodiment, the second step of adjusting the type and amount of the slag composition adjusting material to adjust the composition of the slag

produced in the melting step to have a slag basicity in the range of 0.5 to 2.0 and an Al_2O_3 concentration in the range of 10 to 25 mass %.

[0051] As can be seen from FIG. 2, the slag component contained in the reduced iron has a high FeO concentration, which is effective from the viewpoint of lowering the melting temperature of the slag and thus securing the fluidity of the slag. However, the Fe yield would decrease as it is. Thus, it is preferable to reduce the slag separated by melting, so that the total iron concentration (% $T.Fe$) in the slag becomes 20 mass % or less, more preferably, 10 mass % or less, and further preferably, 5 mass % or less. As a reducing means, any one of the following can be used either alone or in combination: i) supplying a solid containing at least one reducing material selected from the group consisting of a C , Al , and Si to the produced and risen slag, ii) using a gas containing at least one reducing gas selected from the group consisting of CO , H_2 , and hydrocarbon as the gas supplied to the molten iron, and iii) increasing the unit consumption of the reducing material at a time of producing the reduced iron.

[0052] The fourth embodiment of the present invention has been obtained from the above studies, in which, in addition to any one of the first to third embodiments, the melting step includes a fourth step of supplying one or more reducing solids and/or one or more reducing gases. Further, the fifth embodiment of the present invention includes, in addition to the fourth embodiment, adjusting a type and amount of the reducing solid(s) and/or reducing gas(es) to be supplied such that the composition of the slag produced in the melting step has a total iron concentration (% $T.Fe$) of 20 mass % or less.

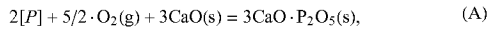
Third Step

[0053] To reduce the temperature drop of the slag produced when the reduced iron is melted in the induction melting furnace, it is preferable to provide a third step of supplying heat to the slag by disposing a heat source above the furnace. The heat source may be, but is not limited to, any means capable of directly heating the slag, such as burner heating, electric heating involving using an electrode, and induction heating performed by immersing a conductor in the slag, and a plurality of such means may also be used in combination. The burner heating may be performed with either a liquid fuel such as heavy oil, or a gas fuel such as CO , H_2 , or hydrocarbon, or a combination of such fuels. The conductor to be immersed in the slag may be any object that generates heat when an induced current is passed there-through. However, from a cost perspective, it is possible to retain an iron rod, a carbon rod, or the like while immersing it in the slag, or add particles of reduced iron produced to achieve bulk density equal to the density of the produced slag from above, and cause the particles to be retained in the slag.

Sixth Embodiment

[0054] Direct reduced iron, which is a raw material, may contain phosphorus as an impurity, and it is preferable to remove the phosphorus from the molten iron. It may also be preferable to add a desired component to the molten iron in some cases. The sixth embodiment of the present invention has been developed from such demand.

[0055] A dephosphorization reaction requires an oxygen source and a CaO source as represented by following Formula (A).



[0056] where [P] represents phosphorus in the molten iron. For example, as a method of removing phosphorus as impurities from the molten iron, it is possible to supply an oxygen source and a CaO source to the molten iron obtained in the melting step or to the molten iron from which the slag has been removed in the slag removal step.

[0057] As an oxygen source for dephosphorization, a pure oxygen gas is typically used. However, considering that a dephosphorization reaction is an exothermic reaction and it is therefore advantageous to perform dephosphorization at a low temperature, the inventors have concluded that it is advantageous to lower the temperature of the molten iron within the range where it does not adversely affect the process.

[0058] As a result of the studies, the inventors have found that it is possible to achieve sufficient dephosphorization while cooling the molten iron by supplying air or an iron oxide source such as iron ore or mill scale as the oxygen source. When air is used, heat removal takes place as sensible heat of a nitrogen gas contained in the air, so that a cooling effect is obtained compared to a case where a pure oxygen gas is used. Meanwhile, when an iron oxide source is used, the iron oxide source is reduced to form metallic Fe or absorption of heat occurs as a molten slag is formed in the form of iron oxide so that a cooling effect is obtained compared to a case where a pure oxygen gas is used.

[0059] Next, the use of limestone as a CaO source can cool the molten iron because the calcium carbonate contained in the limestone absorbs heat as it decomposes into CaO and CO₂. A similar cooling effect can be achieved by supplying carbonate, such as raw dolomite. However, if the proportion of CaO in an auxiliary material is low, the amount of the auxiliary material to be added will increase, causing an operational problem such as an increased amount of slag produced and a longer time required for adding the auxiliary material. Thus, it is preferable to adjust the type and amount of the auxiliary material to be added by considering the required cooling effect and stable operation.

[0060] It is preferable to adjust the supply rate of a pure oxygen or air and the height of a top-blowing lance in accordance with the operation condition of the dephosphorization process because the behavior of the occurrence of spitting differs depending on the height of a freeboard (the height from the position of the upper surface of the molten iron to the position of the upper end of a vessel) of a vessel in which dephosphorization is performed and the nozzle shape of the lance. It is also preferable to blow an inert gas into the molten iron for agitation, especially by providing a porous plug or an injection lance. The slag basicity is preferably in the range of 1.5 to 4.0 and is adjusted based on the amount of the slag containing a large amount of SiO₂ which is carried over to the slag removal step and the type and amount of CaO source added. It is also possible to add a SiO₂ source such as silica stone or ferrosilicon, and a CaO source such as quicklime, as appropriate.

[0061] When the slag basicity is low, the amount of phosphorus removed in dephosphorization will be small. Meanwhile, when the slag basicity is high, a part of the slag will solidify and be attached to a refractory when the temperature of the molten iron drops and cause the slag to be difficult to remove after dephosphorization, causing a problem such that an abnormal reaction may occur when the molten iron is charged in the next process, or the residual slag may be mixed into the produced slag to fail to achieve the desired composition. Since such a dephosphorization process using air produces a large amount of high-temperature exhaust gas, it is also possible to recover the exhaust heat using a boiler, for example.

[0062] Further, the molten iron obtained in the above embodiment may be refined as it is to obtain the components necessary for the next step, resulting in molten steel. Alternatively, the molten iron may be solidified once in a mold box to produce solid iron, and after transporting the solid iron to the place where it is demanded, the solid iron may be remelted and refined into molten steel. The former process can eliminate the solidification and remelting steps, which is energy efficient. However, it will require consecutive installation of a reduced iron production plant, an induction melting furnace, and a refining apparatus. Therefore, if these components are installed in the existing steel mill, the available installation space will be limited. Alternatively, new construction of these apparatuses will require a huge amount of money, and the existing apparatuses cannot be used. In the latter process, it is possible to separate a reduced iron production plant, an induction melting furnace, and a solidification apparatus from a remelting apparatus and a refining apparatus. For example, it is possible to perform a process from the production of reduced iron to solidification in an iron-ore-producing country, and to transport the resulting solidified iron to a place where it is demanded, and then to remelt and refine the solidified iron. In such a case, it is possible not only to utilize the existing refining apparatus but also to transport solidified iron with a weight from which the weight of gangue contained in the iron ore has been reduced. This results in lower transport costs. Which of the process configurations is to be selected may be appropriately determined by considering, for example, the locations of the business facilities and the owned facilities.

[0063] The sixth embodiment of the present invention has been obtained from the above studies and further includes, in addition to any one of the first to fifth embodiments, a refining step of refining the molten iron obtained in the melting step. The refining step is preferably performed after the slag removal step.

Seventh Embodiment

[0064] A seventh embodiment of the present invention is directed to obtaining solid iron by solidifying the molten iron obtained by the method for melting direct reduced iron according to any one of the first to sixth embodiments. It is preferable that the total iron concentration T.Fe in the solid iron be 93 mass % or more and that the total content of oxide components other than Fe be 3 mass % or less. Although the shape and size of a casting mold used to solidify the molten iron into the solid iron are not limited to particular ones, it is preferable that the molten iron be solidified into particles with a size in the range of 10 to 100 mm, considering the following cargo handling, packing, transport, and the supply to a facility where it is to be used, for example.

Eighth Embodiment

[0065] An eighth embodiment of the present invention is directed to utilizing a slag, which is a by-product, as a material for civil engineering and construction. That is, the eighth embodiment further includes a cooling-solidifying step of cooling and solidifying the slag removed in the slag removal step of the above first embodiment for use as a raw material for civil engineering and construction. The cooled and solidified slag has a basicity in the above range and various particle size distributions depending on the cooling method used, and may be used as a material with its properties utilized by performing an additional particle size control process, such as crushing or classification, as appropriate. For example, when subjected to granulation in water, the removed slag is transformed into a fine glass-like form with a specific surface area of 0.35 m²/g or more but less than 0.50 m²/g. The resulting material can be used as a cement raw material (binder). Meanwhile, when the removed slag is slowly cooled in the atmosphere and then subjected to particle size control based on the intended use, the resulting material can be used as a roadbed material or concrete aggregate. As described above, the cooling solidification method may be appropriately selected by a business operator in accordance with the intended use of the removed slag.

Ninth Embodiment

[0066] A ninth embodiment of the present invention is configured as a system for melting direct reduced iron, which can be preferably applied to any of the first to eighth embodiments. This embodiment includes a direct reduction furnace that is configured to obtain a direct reduced iron by bringing iron ore or a mixture of iron ore and a composition adjusting material into contact with a reducing material under heating, an induction melting furnace that is configured to obtain a molten iron by melting the direct reduced iron, and a slag removal mechanism that is configured to remove slag produced in the induction melting furnace to an outside of the melting furnace. As the direct reduction furnace, there may be used a furnace of, for example, shaft furnace-type, fluidized bed-type, and rotary hearth furnace-type. Many of these types are those with which a raw material(s) can be continuously supplied, and a reduced iron can thus be continuously produced. Meanwhile, since induction melting furnaces are configured to carry out batch-type processes, employing a facility where multiple induction melting furnaces are combined with one reduced iron production plant shall make it possible for reduced iron to be supplied to another induction melting furnace while melting the reduced iron in one induction melting furnace, whereby a process waiting time can be reduced. Thus, it is preferable when there are provided two or more induction melting furnaces with respect to one direct reduction furnace. As the slag removal mechanism, there may be provided a slag removal port through which slag can be removed after being overflowed from the upper portion of the induction melting furnace, a tilting mechanism of the furnace body, and/or a slag dragger for dragging out the slag.

[0067] It is preferred that the direct reduction furnace and the induction melting furnace be arranged in proximity to each other so that the direct reduced iron that is to be melted in the induction melting furnace can be charged into the induction melting furnace at a temperature of a range from

a temperature after finishing the direct reduction step to atmospheric temperature. It is preferred that the direct reduced iron be transported to the induction melting furnace while having its temperature drop suppressed.

[0068] As the mechanism for blowing a gas into the molten iron in the induction melting furnace, a gas supplying nozzle may be provided on the furnace bottom and the furnace wall. It may also be that the gas is blown from an immersion lance. A hopper may be provided on the upper portion of the induction melting furnace for the purpose of adding the slag composition adjusting material. Further, there may be installed the heat source described in the third step for the purpose of heating the slag.

EXAMPLES

Example 1

[0069] A mixture of an iron ore having the composition shown in Table 3, quicklime as a composition adjusting material, and a carbon material powder as a reducing material was charged into a rotary hearth furnace yielding a production volume of a scale of 5 ton/hr. Next, a direct reduced iron was produced by performing a reduction treatment where the amounts of a fuel gas and oxygen supplied to a heating burner as well as a ratio therebetween were controlled so that a treatment temperature would be in a range of 1,000° C.±20° C., and an oxygen partial pressure P_{O2} would be in a range of −14 to −15 in terms of logP_{O2}. The amount of quicklime added was determined so that a basicity (ratio of (% CaO)/(% SiO₂) on mass basis) would be about 1. With this facility (rotary hearth furnace), operating conditions were determined so that a time period from charge to discharge would be 90 min, and temperature measurement and gas composition analysis were conducted at a site where the sample charged was present at a time point of 45 min.

TABLE 3

	Ingredient Composition (mass %)			
	T•Fe	SiO ₂	Al ₂ O ₃	P ₂ O ₅
Iron Ore 4	63	3.5	1.4	0.2

[0070] An infrared gas analyzer was used to measure the concentrations of carbon monoxide (CO) and carbon dioxide (CO₂) in the gas collected, where the oxygen partial pressure P_{O2} was calculated from a measured value of a CO/CO₂ ratio as a ratio of each concentration (partial pressure), using following Formulae.

$$2\text{CO}_2(\text{g}) = 2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \quad (\text{B})$$

$$\Delta G^\circ = 134300 - 40.74 \times T(\text{cal/mol}) \quad (3)$$

$$K = \exp(-\Delta G^\circ/RT) = (P_{\text{CO}}/P_{\text{CO}_2})^2 \cdot P_{\text{O}_2} \quad (4)$$

[0071] Here, T represents a reaction temperature (K), K represents an equilibrium constant in the chemical reaction of Formula (B), R represents a gas constant (cal/(K•mol)), and P_{CO} and P_{CO2} respectively represent partial pressures of CO and CO₂ in the gas composition analysis.

[0072] The direct reduced iron obtained through this treatment was then cooled to about 25° C. before being subjected to analysis; as a result, the direct reduced iron was confirmed to have a composition corresponding to the composition of the reduced iron C shown in Table 2.

[0073] The reduced iron that had been produced as above and cooled to room temperature was added into an induction melting furnace having an inner diameter of 0.9 m and a height of 1.8 m from the furnace bottom to the lower end of a tapping gutter and containing 0.5 ton of molten hot metal, such that the reduced iron did not overflow the furnace body. After confirming that the melting had progressed and the height of the layer accumulated in the furnace had decreased, the reduced iron was repeatedly added from above a hopper provided in the upper portion of the furnace until the total amount of the added reduced iron reached 5.0 tons. Six bottom-blowing nozzles are provided at equal intervals at the bottom of the furnace at positions corresponding to a PCD (Pitch Circle Diameter) of 0.3 m and 0.6 m, respectively, to have a configuration that allows gases to be supplied to any combination of nozzles at the same flow rate via a gas header. A hopper capable of supplying an auxiliary material is provided above the furnace so that the auxiliary material can be supplied in units of 10 kg at a given timing of a process. The temperature of the molten iron in

the furnace was measured as appropriate and adjusted to 1600±20° C. by controlling the power of the induction melting furnace or the supply rate of each of the reduced iron and the auxiliary material.

[0074] A melting process was performed by changing the flow rate of gas supplied from the nozzles or the type and amount of the auxiliary material added. Quicklime, which was obtained by roasting limestone at a high temperature to remove CO₂ therefrom and had a CaO concentration of nearly 100 mass %, was used as an auxiliary material. Silica stone was obtained by crushing stone collected at a crushing site. It had a SiO₂ concentration of about 98 mass % and also contained small amounts of Al₂O₃ and MgO. Bauxite was obtained by crushing ore imported as a raw material for Al smelting. It had an Al₂O₃ concentration of about 50 mass % and also contained crystal water, SiO₂, TiO₂, etc. as the remaining impurities. After the process, the furnace body was tilted to completely remove the molten iron from the furnace, and the tapped molten iron was weighed. In addition, slag was collected to be crushed into particles of 53 μm or less and then subjected to a chemical analysis. Tables 4-1 and 4-2 show the values obtained, along with the conditions of the process. For comparison, a process was also performed under a condition that no gas was supplied from the bottom of the furnace.

TABLE 4-1

Auxiliary Material Type: Amount		Gas Supply Conditions			Molten Iron						
		Nozzle		Gas	Supply Amount	Tapped Amount	H	Slag Composition[mass %]			
								T•Fe	SiO ₂	Al ₂ O ₃	CaO
No.	[kg]	Position	Position	Type	Nm ³ /min	t	m	T•Fe	SiO ₂	Al ₂ O ₃	CaO
1	—	Furnace Bottom	1	Ar	0.10	3.59	0.72	45.2	13.1	5.0	13.1
2	—	Furnace Bottom	1	Ar	0.25	3.58	0.72	45.4	13.0	4.9	13.0
3	—	Furnace Bottom	3	Ar	0.10	3.58	0.72	45.4	13.0	4.9	13.0
4	—	Furnace Bottom	3	Ar	0.25	3.58	0.72	45.4	13.0	4.9	13.0
5	—	Furnace Bottom	6	Ar	0.10	3.57	0.72	45.5	13.0	4.9	13.0
6	—	Furnace Bottom	6	Ar	0.25	3.57	0.72	45.5	13.0	4.9	13.0
7	Quicklime: 50	Furnace Bottom	3	Ar	0.25	3.56	0.72	44.7	12.5	4.8	15.0
8	Quicklime: 150	Furnace Bottom	3	Ar	0.25	3.52	0.71	43.4	11.6	4.4	18.6
9	Quicklime: 250	Furnace Bottom	3	Ar	0.25	3.49	0.70	41.9	11.0	4.2	21.9
10	Silicestone: 100	Furnace Bottom	3	Ar	0.25	3.53	0.71	44.2	16.8	4.6	12.0
11	Silicestone: 250	Furnace Bottom	3	Ar	0.25	3.49	0.70	42.0	21.8	4.2	11.0
12	Bauxite: 140	Furnace Bottom	3	Ar	0.25	3.52	0.71	44.9	12.1	8.0	12.1
13	—	Furnace Bottom	1	Ar	0.70	3.49	0.70	45.0	13.2	5.0	13.2
14	—	Furnace Bottom	3	Ar	2.1	3.26	0.66	45.2	13.1	5.0	13.1
15	—	Furnace Bottom	6	Ar	4.2	3.01	0.61	45.4	13.0	4.9	13.0
16	Quicklime: 300	Furnace Bottom	3	Ar	0.25	3.48	0.70	41.4	10.6	4.0	23.4
17	Silicestone: 300	Furnace Bottom	3	Ar	0.25	3.48	0.70	41.5	23.2	4.1	10.7
18	Bauxite: 150	Furnace Bottom	3	Ar	0.25	3.52	0.71	45.0	1.21	8.2	12.1
19	Quicklime: 200, Silicestone: 200	Furnace Bottom	3	Ar	0.25	3.46	0.70	40.2	18.0	3.8	18.2
20	—	—	—	—	—	1.79	0.36	45.5	13.0	4.9	13.0

TABLE 4-2

Determination							
					Processability		
No	Right-Hand Side of Formula (2)	Formula (2)	C/S Range	Al ₂ O ₃ Concentration	Slag Fluidity	Spitting of Molten Iron	Remarks
1	0.20	Satisfied	Within Range	Within Range	Good	Small	Invention Example
2	0.37	Satisfied	Within Range	Within Range	Good	Small	Invention Example
3	0.10	Satisfied	Within Range	Within Range	Good	Small	Invention Example
4	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example
5	0.06	Satisfied	Within Range	Within Range	Good	Small	Invention Example

TABLE 4-2-continued

Determination							
No	Right-Hand Side of Formula (2)	Formula (2)	Processability				
			C/S Range	Al ₂ O ₃ Concentration	Slag Fluidity	Spitting of Molten Iron	Remarks
6	0.11	Satisfied	Within Range	Within Range	Good	Small	Invention Example
7	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example
8	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example
9	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example
10	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example
11	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example
12	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example
13	0.74	Not Satisfied	Within Range	Within Range	Good	Large	Invention Example
14	0.74	Not Satisfied	Within Range	Within Range	Good	Large	Invention Example
15	0.74	Not Satisfied	Within Range	Within Range	Good	Large	Invention Example
16	0.18	Satisfied	Without Range	Within Range	Fair	Small	Invention Example
17	0.18	Satisfied	Without Range	Within Range	Fair	Small	Invention Example
18	0.18	Satisfied	Within Range	Without Range	Fair	Small	Invention Example
19	0.18	Satisfied	Within Range	Without Range	Fair	Small	Invention Example
20	—	—	Within Range	Within Range	Poor	Small	Comparative Example

[0075] The fluidity of each slag in Table 4-2 was determined by observing the surface of the slag in the furnace through an observation window provided above the furnace. Specifically, a state in which the entire surface of the slag was red hot and high-temperature slag was always circulating is indicated as “Good,” a state in which the slag surface was at least partially black in a solid state but the solid state slag was always moving on the surface is indicated as “Fair,” and a state in which the entire slag surface was black except that cracked portions were red hot and the slag was thus stagnated is indicated as “Poor.” Note that under the condition that the fluidity of the slag was “Good,” it took 90 minutes or less from the time when 5.0 tons of reduced iron was added to the time when the reduced iron was completely melted and the furnace body was tilted for tapping. Meanwhile, under the condition that the fluidity of the slag was “Fair,” it took more than 90 minutes. The difference is due to the fact that part of the slag was solidified and it thus took some time to transfer the slag to a tapping port in the slag removal step. Under the condition that the fluidity of the slag was “Poor,” even when the reduced iron and auxiliary material were additionally charged, they accumulated on the surface of the solidified slag. As a result, the reduced iron and auxiliary material did not penetrate into the inside of the molten iron, and only the reduced iron charged in the initial stage could be melted.

[0076] The fluidity of the slag was evaluated as “Good” or “Fair” in Test Nos. 1 through 19, while it was evaluated as “Poor” in Test No. 20. This is considered due to the fact that in Test No. 20 no gas was supplied from the bottom of the furnace, sufficient heat was not supplied to the slag from the molten iron and the slag solidified. Meanwhile, in each of Test Nos. 1 through 19, where gas was supplied, the fluidity of the slag was maintained regardless of the slag composition, and thus the slag was removed. Note that, as is clear from Tables 4-1 and 4-2, the fluidity of the slag was particularly excellent when Formula (1) and Formula (2) above were satisfied, the slag basicity C/S was in the range of 0.5 to 2.0, and the Al₂O₃ concentration in the slag as converted to that on a CaO—SiO₂—Al₂O₃ ternary phase diagram was in the range of 10 mass % to 25 mass %.

[0077] In Test Nos. 13 to 15, the amount of the gas supplied from the bottom of the furnace was too large, causing the blow-by phenomenon described above, so that the spitting of the molten iron was significant. As a result, the amount of the tapped molten metal was smaller than that of the other invention examples. In Test No. 16, since an excessive amount of quicklime was added, the slag basicity C/S was over 2.0, and the melting temperature of the slag thus increased, resulting in lower fluidity and a longer slag removal time. In Test No. 17, since an excessive amount of silica stone was added, the slag basicity C/S was less than 0.5, and the viscosity of the slag thus increased, resulting in lower fluidity and a longer slag removal time. In Test No. 18, since an excessive amount of bauxite was added, the Al₂O₃ concentration as converted to that on a CaO—SiO₂—Al₂O₃ ternary phase diagram was over 25 mass %, and the melting temperature of the slag thus increased, resulting in lower fluidity and a longer slag removal time. In Test No. 19, since excessive amounts of quicklime and silica stone were added, the Al₂O₃ concentration as converted to that on a CaO—SiO₂—Al₂O₃ ternary phase diagram was diluted to less than 5 mass %, and the melting temperature of the slag thus increased, resulting in lower fluidity and a longer slag removal time.

[0078] The auxiliary material is not limited to quicklime, silica stone, or bauxite, and it was confirmed that there is no problem in using the other materials described above to control the composition of the slag. It was also found that it is more preferable to add the auxiliary material in stages from the start of the melting process to the completion of the melting process. This is due to the fact that if a large amount of auxiliary material is added immediately after the start of the melting process, the contact between the reduced iron or between the reduced iron and the hot metal will be hindered, resulting in a lower efficiency of the induction heating and a longer melting time. By the time the melting is completed, slag has already been produced, or solidified depending on its composition. Therefore, even if the auxiliary material is added at this point, it will only be placed on the solidified slag, and will not contribute to lowering the melting temperature of the slag.

[0079] The reduced iron could be melted without the presence of hot metal. However, in the presence of hot metal, heat would be supplied from the inductively heated hot metal to the solid reduced iron, which in turn would reduce the time required to melt the reduced iron. Thus, hot metal is preferably present. In particular, it is effective to provide about 5 mass % or more of hot metal with respect to the reduced iron to be charged. Meanwhile, a larger amount of hot metal would increase the ratio of hot metal to the volume of the induction melting furnace, which in turn would decrease the amount of the reduced iron that can be melted. It is therefore preferable to set the ratio of hot metal to the reduced iron to be charged to 70 mass % or less. In addition, the hot metal may also be provided by newly melting scraps with a high bulk density or a large lump of scull, or by leaving part of the molten iron melted in the previous melting process in the furnace.

[0080] Note that the process was performed by changing the nozzle position and a combination of nozzles. When the amount of the gas supplied per nozzle satisfied above Formula (2), the slag fluidity was secured and the spitting of molten iron was small regardless of the nozzle position or a combination of nozzles, while when the amount of the gas supplied per nozzle did not satisfy above Formula (2), the spitting of molten iron was increased, resulting in lower metal yields.

[0081] It was confirmed that the nozzle position may be not only on the bottom of the furnace but also on the side surface of the furnace. However, if the distance h from the

nozzle provided on the side surface of the furnace to the bottom of the furnace is long, that is, if H is small, the gas supply rate that satisfies Formula (2) will decrease, which in turn would make it difficult to effectively transfer the heat of the molten iron to the slag. Thus, the nozzle position is preferably located at or near the bottom of the furnace as much as possible.

Example 2

[0082] A FeO component contained in the produced slag was reduced using the same induction melting furnace as the direct reduced iron produced by the same method as that in Example 1. A gas reducing material such as CO, H₂, and CH₄ was supplied from bottom-blowing nozzles, and the supply time of each gas reducing material was set constant: 90 minutes. In addition, solid C, metallic Al, and metallic Si as solid reducing materials were each added to the produced slag from the above. Herein, when metallic Al and metallic Si were used for the reduction, the basicity and Al₂O₃ concentration of the produced slag changed. Thus, quicklime or silica stone was added as an auxiliary material to adjust the basicity and Al₂O₃ concentration of the slag. After the process, the furnace body was tilted to completely remove the molten iron from the furnace, and the tapped molten iron was weighed. In addition, the slag was collected to be crushed into particles of 53 μm or less and then subjected to a chemical analysis. Tables 5-1 and 5-2 show the obtained values along with the conditions of the process.

TABLE 5-1

Auxiliary Material Type: Amount		Gas Supply Conditions			Molten Iron						
		Nozzle		Gas	Supply Amount	Tapped Amount	H	Slag Composition [mass %]			
								Position	Number	Type	Nm ³ /min
No.	[kg]										
21	—	Furnace Bottom	3	CO	0.25	3.63	0.73	44.3	13.4	5.1	13.4
22	—	Furnace Bottom	3	H ₂	0.25	3.63	0.73	44.3	13.4	5.1	13.4
23	—	Furnace Bottom	3	CH ₄	0.25	3.78	0.76	40.3	15.1	5.7	15.1
24	—	Furnace Bottom	3	CH ₄	0.50	3.98	0.80	33.3	17.9	6.8	17.9
25	—	Furnace Bottom	3	CH ₄	0.80	4.24	0.85	19.9	23.3	8.8	23.3
26	—	Furnace Bottom	3	CH ₄	1.00	4.36	0.88	9.4	27.5	10.4	27.5
27	—	Furnace Bottom	3	CH	1.20	4.41	0.89	4.6	29.4	11.2	29.4
28	Solid C: 100	Furnace Bottom	3	Ar	0.25	4.00	0.81	32.7	18.1	6.9	18.1
29	Metallic Al: 50, Quicklime: 30, Silicastone: 30	Furnace Bottom	3	Ar	0.25	3.72	0.75	38.7	14.8	9.8	14.8
30	Metallic Si: 50, Quicklime: 100	Furnace Bottom	3	Ar	0.25	3.76	0.76	36.6	18.6	5.0	18.5

TABLE 5-2

Determination							
					Processability		
No.	Right-Hand Side of Formula (2)	Formula (2)	C/S Range	Al ₂ O ₃ Concentration	Slag Fluidity	Spitting of Molten Iron	Remarks
21	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example
22	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example
23	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example
24	0.28	Satisfied	Within Range	Within Range	Good	Small	Invention Example
25	0.39	Satisfied	Within Range	Within Range	Good	Small	Invention Example
26	0.45	Satisfied	Within Range	Within Range	Good	Small	Invention Example
27	0.51	Satisfied	Within Range	Within Range	Good	Small	Invention Example

TABLE 5-2-continued

Determination							
No.	Right-Hand	Formula (2)	C/S Range	Al ₂ O ₃ Concentration	Processability		Remarks
	Side of Formula (2)				Slag Fluidity	Spitting of Molten Iron	
28	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example
29	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example
30	0.18	Satisfied	Within Range	Within Range	Good	Small	Invention Example

[0083] In each of Test Nos. 21 to 27 where the reducing gas was supplied, the amount of tapped metal increased, and the T.Fe concentration in the slag decreased compared to Test No. 4 of Example 1. This indicates that the gas reducing material effectively contributed to the reduction of FeO in the produced slag. In addition, when comparing Test Nos. 23 to 27, it is found that increasing the supply amount of the reducing gas (CH₄) reduced T.Fe in the slag and thus increased the amount of tapped metal.

[0084] In Test Nos. 28 to 30 where the solid reducing material was added, the amount of tapped metal also increased and the T.Fe concentration in the slag decreased compared to Test No. 4 of Example 1. This indicates that the solid reducing materials, in the same as the reducing gas, effectively contributed to the reduction of FeO in the produced slag.

[0085] Various evaluations were conducted by changing the type, combination, amount, and the timing of reducing material addition, for example. In each case, the amount of tapped metal increased and the T.Fe concentration in the slag decreased, compared to a condition that no reducing material was added. In order to achieve the objective of producing molten iron of the present technology, it is preferable to reduce the slag by supplying a reducing material thereto to improve the iron yield so that the T.Fe concentration in the slag becomes 20 mass % or less, more preferably, 10 mass % or less, and further preferably, 5 mass % or less. It should be noted that the use efficiency of the supplied reducing material decreases as the reduction process proceeds. Thus, it is important to determine a method for performing the process by considering the ingredient composition and costs of reduced iron, auxiliary material, and reducing material.

Example 3

[0086] There was performed a process in which the high-temperature direct reduced iron produced by the rotary hearth furnace described in Example 1 was directly charged into an induction melting furnace and melted therein. A direct reduced iron whose temperature was approximately 1,000° C. when exiting the rotary hearth furnace was directly charged into an induction melting furnace similar to those used in Examples 1 and 2 and melted therein. Here, the temperature of the reduced iron dropped during its transport before charging, which caused the temperature of the reduced iron to reach about 900° C. immediately after charging. Even the reduced iron in the high-temperature state was able to be heated by induction without any problem, and a time required for temperature rise was able to be shortened by 30 minutes or more as compared to when treating a reduced iron of an equivalent composition under equivalent treatment conditions. Here, even in the case of melting the reduced iron that had been directly charged in a heated condition, the fluidity of the slag was ensured when

the aforementioned conditions for gas blowing and slag compositions were satisfied, which made it possible to carry out the process in a stable manner. Further, if the aforementioned conditions for slag basicity and Al₂O₃ concentration were already met at the time of producing the reduced iron, only gas supply needed to be performed under the conditions satisfying Formula (2) at the time of melting the reduced iron in the induction melting furnace, and there was no need to add the auxiliary material at that time.

Example 4

[0087] The molten iron obtained in each of Examples 1 and 2 was subjected to temperature adjustment and then transferred to a pot-shaped vessel. Among the slag produced due to the gangue contained in the reduced iron during the melting of the reduced iron in the induction melting furnace, approximately 10 kg of slag per 1 ton of molten iron was transferred to the pot-shaped vessel together with the molten iron, and the rest of the slag was transferred to another slag vessel. The pot-shaped vessel was transferred to a dephosphorization facility to perform dephosphorization while changing the types and amounts of an oxygen source and a lime source supplied. The dephosphorization facility included a gas top-blowing lance, an auxiliary material feeding hopper, and a bottom-blowing porous plug. The gas top-blowing lance was capable of supplying a gas containing pure oxygen or air at a rate of approximately 1 Nm³/minute per 1 ton of molten iron. Three auxiliary material feeding hoppers, each filled with iron ore, quicklime (CaO), and calcium carbonate (CaCO₃), can feed them at a rate of approximately 10 kg/minute. The bottom-blowing porous plug can supply a gas. In this example, a pure Ar gas was supplied at a rate of approximately 0.1 Nm³/minute per 1 ton of molten iron.

[0088] The melting temperature in the induction melting furnace was adjusted to allow the temperature of the molten iron before dephosphorization to be approximately 1590° C. “Before dephosphorization” refers to the time before the gas top-blowing lance is lowered, while “after dephosphorization” denotes the time when the gas top-blowing lance has been completely raised after the dephosphorization. At each timing, temperature measurements and sampling were conducted using a substance. The obtained samples were cut and polished and subjected to an emission spectrochemical analysis to evaluate the C concentration [C] and the P concentration [P] in the molten iron from calibration curves determined in advance. It was possible to measure the solidifying temperature of the molten metal at the timing when the temperature measurement and sampling were performed using the substance, and the solidifying temperature T_m of the molten iron subjected to the dephosphorization was actually measured.

[0089] The start of dephosphorization was defined as when the gas top-blowing lance started to be lowered. After

the top-blowing lance reached a predetermined height, the supply of an oxygen gas source and the addition of auxiliary material were started. The dephosphorization was terminated when the supply of predetermined amounts of oxygen gas source and auxiliary material was completed and the top-blowing lance was raised to a standby position. The duration of the period was determined as a processing time t_f (minutes).

[0090] After the dephosphorization, the pot-shaped vessel was tilted to remove the slag on the molten iron with a slag dragger. Part of the removed slag was taken and subjected to a chemical analysis. The pot was lifted and tilted using a crane to transfer the molten iron to the tundish. The molten iron was caused to flow down from the tundish to collide with a surface plate, and the resulting molten iron droplets were dropped into the cooling water tank and solidified to produce grained iron. The grain sizes of the obtained grained iron ranged from 0.1 to 30 mm. The grain size distributions were: +0.1 mm to -1 mm: 17.2 mass %, +1 mm to -10 mm: 31.3 mass %, +10 mm to -20 mm: 38.8 mass %, and +20 mm to -30 mm: 12.7 mass %. Herein, “+N to -M” means particles on a sieve with an opening of N to particles that have passed through a sieve with an opening of M.

[0091] The temperature T_f of the molten iron after dephosphorization was adjusted to be lower than the temperature T_i of the molten iron before dephosphorization while the slag basicity C/S was adjusted to be in the range of 1.5 to 4.0, so that the temperature T_f of the molten iron after dephosphorization was adjusted to be higher than the solidifying temperature T_m of the molten iron by 20° C. or more. As a result, a P concentration $[P]_i$ of about 0.12 mass % in the molten iron before dephosphorization decreased to a P concentration $[P]_f$ of 0.02 to 0.04 mass % in the molten iron after dephosphorization. In addition, solid iron was produced without affecting the productivity of grained iron.

[0092] When the molten iron produced in each invention example according to Examples 1 and 2, including the grained iron produced as above, was solidified, it was confirmed that solid iron having a total iron content T.Fe of 93 mass % or more and a total content of oxide components other than Fe of 3 mass % or less was obtained regardless of the size or shape of the casting mold used. Although the size and shape of the casting mold may be changed in accordance with the intended use of the solid iron to be required, it is preferable to solidify the molten iron into particles with a size in the range of 10 mm to 100 mm, considering the following cargo handling, packing, transport, and the supply to a facility where it is to be used, for example.

[0093] The slag produced in each invention example according to Examples 1 and 2 has the fluidity required for the slag to be removed. The slag basicity C/S is in the range of 0.5 to 2.0, and the Al_2O_3 concentration in the slag as converted to that on a $CaO-SiO_2-Al_2O_3$ ternary phase diagram is in the range of 10 to 25 mass %. When subjected to granulation in water, the molten slag was transformed into a fine glass-like form with a specific surface area of 0.35 m^2/g or more but less than 0.50 m^2/g , which can be used as a cement raw material. Meanwhile, when the molten slag is slowly cooled in the atmosphere, a lump of slag with a size of about several hundred mm or less is obtained. Such a lump of slag can be used as a subgrade material or a concrete aggregate by performing appropriate particle size adjustment through crushing and classification.

[0094] In this specification, the unit “t” of a mass represents 10^3 kg, and the unit for heat quantity “cal” is converted into “4.184 J.” In addition, the symbol “N” added to the unit “ Nm^3 ” of a volume represents the standard state of a gas. In

this specification, the standard state of a gas corresponds to 1 atm (=101325 Pa) and 0° C. Symbol [M] in a chemical formula represents that an element M is melted in molten iron or reduced iron.

INDUSTRIAL APPLICABILITY

[0095] The method for melting direct reduced iron of the present invention includes, when melting direct reduced iron in an induction melting furnace, blowing gas into the molten iron to increase the fluidity of a slag, thereby separating the slag from the molten iron; and particularly, melting a high-temperature reduced iron to thereby save energy and improve productivity. Thus, the present invention is industrially advantageous.

1. A method for melting direct reduced iron, comprising:
 - a direct reduction step of obtaining a direct reduced iron by bringing iron ore or a mixture of iron ore and a composition adjusting material into contact with a reducing material under heating;
 - a melting step of melting the direct reduced iron in an induction melting furnace to produce molten iron;
 - a slag removal step of removing slag produced in the melting step to an outside of the induction melting furnace; and,
 - optionally, a refining step of refining the molten iron obtained in the melting step,

wherein

- a charging temperature of the direct reduced iron into the induction melting furnace in the melting step of melting the direct reduced iron is in a range from a temperature after finishing the direct reduction step to atmospheric temperature, and
- the melting step includes a first step of blowing gas into the molten iron for a limited time of or throughout the melting step, and, optionally, one or more steps selected from
 - 1) a second step of adding a slag composition adjusting material,
 - 2) a third step of supplying heat to the slag from a heat source disposed above the induction melting furnace, and
 - 3) a fourth step of supplying one or more reducing solids and/or one or more reducing gases.

2. The method for melting direct reduced iron according to claim 1, wherein

the first step includes, provided that a height H(m) from a position of a gas supply nozzle for blowing the gas into the molten iron to a bath surface of the molten iron is represented by following Formula (1), blowing the gas into the molten iron so as to satisfy following Formula (2):

$$H = 1.27 \times W_{DRI} / (\rho_1 D^2) \times (\% T \cdot Fe)_{DRI} / 100 - h \quad (1)$$

$$H > 0.18 \times (\rho_g Q^2 / \rho_1 N^2 d^2)^{1/3}, \quad (2)$$

where ρ_g represents a density (kg/m^3) of the supplied gas, ρ_l represents a density (kg/m^3) of the molten iron, Q represents a supply rate ($Nm^3/minute$) of the gas, N represents the number (—) of gas supply nozzles, d represents a diameter (m) of each gas supply nozzle, D represents an inner diameter (m) of the induction furnace,

W_{DRI} represents a weight (kg) of reduced iron supplied into the induction furnace,
 (% T.Fe) $_{DRI}$ represents a total iron concentration (mass %) in the reduced iron, and
 h represents a height (m) from a bottom of the induction furnace to the position of the gas supply nozzle.

3. The method for melting direct reduced iron according to claim 1, wherein
 the second step comprises adjusting a type and amount of the slag composition adjusting material added so that the composition of the slag produced in the melting step can achieve a basicity, which is a ratio of a CaO concentration (% CaO) to a SiO₂ concentration (% SiO₂) on a mass basis, in a range of 0.5 to 2.0 and an Al₂O₃ concentration (% Al₂O₃) in a range of 10 to 25 mass %.

4. The method for melting direct reduced iron according to claim 1, wherein
 the fourth step comprises adjusting a type and amount of the reducing solid(s) and/or reducing gas(es) to be supplied so that the composition of the slag produced in the melting step can achieve a total iron concentration (% T.Fe) of 20 mass % or less.

5. A method for producing solid iron, comprising solidifying the molten iron obtained with the method according to claim 1 to obtain solid iron.

6. Solid iron produced by the method according to claim 5, wherein:
 a total iron concentration T.Fe is 93 mass % or more, and
 a total content of oxide components other than Fe is 3 mass % or less.

7. A method for producing a material for civil engineering and construction, comprising:
 a melting step of melting direct reduced iron in an induction melting furnace to obtain molten iron,
 a slag removal step of removing slag produced in the melting step to an outside of the melting furnace, and
 a cooling solidification step of cooling the slag removed in the slag removal step to solidify the slag into a raw material for civil engineering and construction,
 wherein
 a charging temperature of the direct reduced iron into the induction melting furnace in the melting step of melting the direct reduced iron is in a range from a temperature after finishing the direct reduction step to atmospheric temperature, and
 the melting step includes a first step of blowing gas into the molten iron for a limited time of or throughout the melting step, and, optionally, one or more steps selected from
 1) a second step of adding a slag composition adjusting material and
 2) a third step of supplying heat to the slag from a heat source disposed above the induction melting furnace.

8. A material for civil engineering and construction produced with the method according to claim 7, wherein:
 a basicity that is a ratio in mass of a CaO concentration (% CaO) to a SiO₂ concentration (% SiO₂) is in a range of 0.5 to 2.0, and
 an Al₂O₃ concentration (% Al₂O₃) is in a range of 10 to 25 mass %.

9. A system for melting direct reduced iron, comprising:
 a direct reduction furnace that is configured to obtain a direct reduced iron by bringing iron ore or a mixture of iron ore and a composition adjusting material into contact with a reducing material under heating;
 an induction melting furnace that is configured to obtain a molten iron by melting the direct reduced iron; and
 a slag removal mechanism that is configured to discharge a slag generated in the induction melting furnace to an outside of the melting furnace; and
 further optionally comprising:
 a refining facility that is configured to refine the molten iron melted in the induction melting furnace,
 wherein:
 the direct reduced iron that is to be melted in the induction melting furnace is charged into the induction melting furnace at a temperature of a range from a temperature after finishing the direct reduction treatment to atmospheric temperature, the induction melting furnace includes a function of blowing a gas into the molten iron for a limited time of or throughout the melting of the direct reduced iron, and further optionally includes one or more functions selected from:
 1) a function of adding an adjuster for adjusting components of the slag,
 2) a function of supplying heat to the slag from a heat source disposed above the induction melting furnace, and
 3) a function of supplying one or more reducing solids or gases.

10. The system for melting direct reduced iron according to claim 9, wherein the system comprises two or more said induction melting furnaces with respect to one said direct reduction furnace.

11. A method for producing solid iron, comprising solidifying the molten iron obtained with the method according to claim 2 to obtain solid iron.

12. A method for producing solid iron, comprising solidifying the molten iron obtained with the method according to claim 3 to obtain solid iron.

13. A method for producing solid iron, comprising solidifying the molten iron obtained with the method according to claim 4 to obtain solid iron.

14. Solid iron produced by the method according to claim 11, wherein:
 a total iron concentration T.Fe is 93 mass % or more, and
 a total content of oxide components other than Fe is 3 mass % or less.

15. Solid iron produced by the method according to claim 12, wherein:
 a total iron concentration T.Fe is 93 mass % or more, and
 a total content of oxide components other than Fe is 3 mass % or less.

16. Solid iron produced by the method according to claim 13, wherein:
 a total iron concentration T.Fe is 93 mass % or more, and
 a total content of oxide components other than Fe is 3 mass % or less.

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